

Stock Portfolio Analysis (Excel)

Project Documentation

[GitHub Repository](#)

Contents

1	Stock Portfolio Analysis (Excel)	2
2	Data Disclaimer	2
3	Overview	2
4	Project Scope / Analytical Coverage	2
5	Project Results	2
5.1	Transaction & Portfolio Activity	2
5.2	Market Data & Price Monitoring	3
5.3	Stock Screening & Selection	3
5.4	Link to Excel Model	3
6	Stakeholders & Potential Applications	3
7	Workbook Overview	4
7.1	Transaction History	4
7.2	Real Time Market Price	4
7.3	Search Engine	4
8	Vocabulary	5
8.1	Transaction Vocabulary	5
8.2	Portfolio Vocabulary	5
8.3	Performance & Risk Vocabulary	6
8.4	Market & Classification Vocabulary	6
8.5	Sector Definition	6
9	Formulas & Functions	7
9.1	Formulas	7
9.2	Functions	7
9.3	Function Logistics	8
9.3.1	Function 01: Running Cost Basis	8
9.3.2	Function 02: Total Invested	10
9.3.3	Function 03: Amount Invested	10
9.3.4	Function 04: Real Time Value	12
9.3.5	Function 05: Purchase History Tracker	13

1 Stock Portfolio Analysis (Excel)

Portfolio information may be distributed across transaction records, market prices, holdings, and company-level data, making it difficult to obtain a consistent view of portfolio performance and risk.

2 Data Disclaimer

All stock transaction data in this project are entirely illustrative and created solely for demonstration purposes. The transactions do not represent actual personal investments, trading activity, or brokerage records. They are used to demonstrate the analytical model, calculation logic, and Excel capabilities of the project.

3 Overview

Portfolio information may be distributed across transaction records, market prices, holdings, and company-level data, making it difficult to obtain a consistent view of portfolio performance and risk. This project is an Excel-based Stock Portfolio Analysis Model that transforms transaction and market data into portfolio holdings, cost basis, P&L, performance, allocation, and risk metrics. The project follows an analyst-oriented workflow: defining business questions, structuring data requirements, building analytical logic, calculating KPIs, and communicating findings through dashboards.

4 Project Scope / Analytical Coverage

1. **Transaction Tracking** — Records buy and sell activities, including transaction dates, quantities, prices, and transaction values.
2. **Portfolio Holdings** — Calculates current holdings, position sizes, market values, and portfolio weights.
3. **Cost Basis** — Tracks remaining cost basis based on transaction activity.
4. **P&L Analysis** — Calculates realized and unrealized P&L at the security and portfolio levels.
5. **Performance Analysis** — Evaluates portfolio and security-level returns and benchmark performance.
6. **Portfolio Allocation** — Analyzes exposure across individual securities and sectors.
7. **Risk Analysis** — Evaluates selected portfolio risk metrics, concentration, and exposure.
8. **Dashboard Reporting** — Consolidates key portfolio KPIs and analytical results into an interactive dashboard.
9. **Excel Automation** — Uses Excel formulas, PivotTables, and VBA to automate calculations and repetitive reporting tasks.

5 Project Results

5.1 Transaction & Portfolio Activity

Tracks buy and sell transactions and provides the underlying activity used for portfolio analysis.

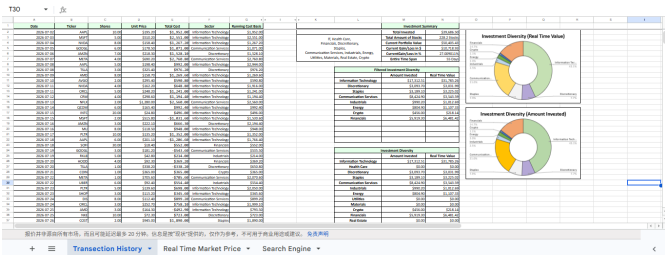


Figure 1: Transaction & Portfolio Activity

5.2 Market Data & Price Monitoring

Provides market price information used to calculate portfolio value, P&L, and performance.

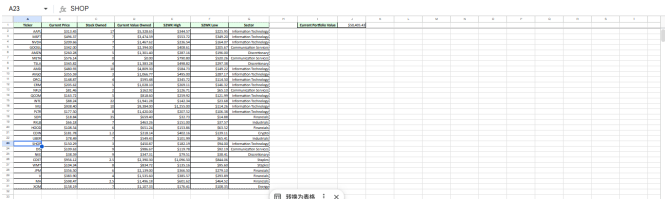


Figure 2: Market Data & Price Monitoring

5.3 Stock Screening & Selection

Screens securities based on selected criteria to support stock analysis and selection.

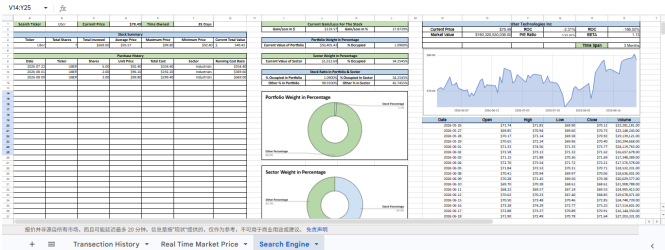


Figure 3: Stock Screening & Selection

5.4 Link to Excel Model

[Stock Analyst Excel Model.xlsx](#)

6 Stakeholders & Potential Applications

The model is designed around different stakeholder information needs and can be adapted to portfolio monitoring, investment reporting, and financial analysis use cases.

Stakeholder	Key Information Needs	Potential Application
Individual Investor	Portfolio value, P&L, return, and allocation	Portfolio Monitoring
Investment Analyst	Security performance, contribution, benchmark, and risk	Investment Research & Performance Analysis
Portfolio Manager	Position size, sector exposure, concentration and risk	Portfolio Review & Risk Monitoring
Decision Maker	Key performance and risk indicators	Management Reporting & Decision Support

Figure 4: Stakeholder & Potential Applications

7 Workbook Overview

The workbook is organized into three main worksheets, moving from transaction-level portfolio information to market data and interactive security-level analysis.

7.1 Transaction History

Provides a historical view of portfolio transactions and summarizes investment activity and portfolio diversification.

1. **Transaction History Chart** — Visualizes historical buying and selling activity over time.
2. **Investment Summary** — Summarizes key investment information and portfolio-level metrics.
3. **Investment Diversity Chart** — Shows how investments are distributed across different securities.
4. **Investment Diversity Diagram** — Provides a visual representation of portfolio diversification.

7.2 Real Time Market Price

Provides the latest available market price for each ticker currently included in the portfolio.

1. **Real-Time Price by Ticker** — Displays the current market price for each portfolio security and provides the price inputs used for portfolio valuation and gain/loss analysis.

7.3 Search Engine

Provides an interactive security-level analysis tool that allows users to search for a specific ticker and review its portfolio position, purchase history, performance, allocation, and market information.

1. **Search Area** — Allows users to select or enter a ticker for detailed analysis.
2. **Stock Summary** — Summarizes key information and current portfolio metrics for the selected security.
3. **Purchase History** — Displays the historical purchase activity for the selected ticker.
4. **Current Gain & Loss Chart** — Visualizes the current gain or loss associated with the selected security.
5. **Portfolio Weight (%) Chart** — Shows the selected security's percentage of total portfolio value.
6. **Portfolio Weight (%) Diagram** — Provides a visual representation of the selected security's contribution to the overall portfolio.

7. **Sector Weight (%) Chart** — Shows portfolio exposure by sector.
8. **Sector Weight (%) Diagram** — Provides a visual representation of sector diversification.
9. **Real-Time Price Summary** — Displays the latest market price and related information for the selected ticker.
10. **Real-Time Price Chart** — Visualizes the selected ticker's market price information.
11. **Real-Time Price Diagram** — Provides an additional visual representation of the selected security's current market price.

8 Vocabulary

8.1 Transaction Vocabulary

Term	Definition / Project Use
Ticker	A unique symbol used to identify a publicly traded stock. Used to identify and track the stock being transacted and to connect transaction, market-price, and stock-summary data.
Date	The date on which a transaction occurs. Used to record the timing of each Buy or Sell transaction and to calculate the holding period of a stock.
Shares	The number of shares bought or sold in a transaction. Used to calculate Total Cost, current shares owned, and Current Value.
Unit Price	The price paid or received for one share in a transaction. Used with Shares to calculate Total Cost and determine cost basis.
Total Cost	The total dollar amount associated with a transaction, calculated as Shares $\times$ Unit Price.
Buy	A transaction in which shares are purchased. Increases shares owned and adds transaction cost to Running Cost Basis.
Sell	A transaction in which shares are sold. Decreases shares owned and reduces Running Cost Basis.
Transaction	A recorded purchase or sale of a financial asset. Each row in Transaction History represents one Buy or Sell transaction.

8.2 Portfolio Vocabulary

Term	Definition / Project Use
Portfolio	A collection of financial assets held by an investor or investment entity.
Holding	An asset or number of shares currently owned in a portfolio.
Position	The amount of a particular security held within a portfolio.
Total Invested	The total amount of capital invested in a stock or portfolio based on the model's investment calculation.
Current Value	Current market value based on current price and quantity held; calculated as Current Price $\times$ Stock Owned.
Market Value	The value of an asset or position based on its current market price.
Portfolio Weight	The percentage of total portfolio market value represented by a position; Stock Current Value $\div$ Current Portfolio Value.

Portfolio Allocation	Distribution of portfolio value across individual assets, sectors, or other categories.
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8.3 Performance & Risk Vocabulary

Term	Definition / Project Use
P&L	Financial gain or loss generated by an investment or portfolio.
Gain/Loss in Dollars	Absolute dollar profit or loss; $\text{Current Value} - \text{Invested Amount}$.
Gain/Loss in Percentage	Gain/loss relative to invested amount; $(\text{Current Value} - \text{Invested Amount}) \div \text{Invested Amount}$.
Realized P&L	Profit or loss resulting from a position that has been sold.
Unrealized P&L	Profit or loss on an asset that is still being held.
Return	Gain or loss from an investment expressed relative to capital invested.
ROC (Rate of Change)	Percentage measure of how much price or value has changed between two points in time.
Beta	Measure of a stock's sensitivity to movements in the overall market.

8.4 Market & Classification Vocabulary

Term	Definition / Project Use
Sector	Broad classification grouping companies based on primary business type.
Industry	More specific business classification describing products or services.
Current Price	Most recent market price available for a stock.
52-Week High	Highest market price reached during the previous 52 weeks.
52-Week Low	Lowest market price reached during the previous 52 weeks.
P/E Ratio	Valuation ratio comparing stock price with earnings per share.
Benchmark	Standard market index or reference used to evaluate investment performance, such as the S&P 500.

8.5 Sector Definition

Sector	Definition
Information Technology (IT)	Software, hardware, semiconductors, technology services, and related products.
Health Care	Pharmaceuticals, biotechnology, medical devices, healthcare services, and related products.
Financials	Banking, insurance, investment, lending, and other financial services.
Consumer Discretionary	Non-essential consumer goods and services, including retail, automobiles, travel, entertainment, and leisure.
Consumer Staples	Essential consumer products such as food, beverages, household goods, and personal care products.
Communication Services	Communication and related consumer services.
Industrials	Manufacturing, machinery, transportation, aerospace, construction, and industrial services.

Energy	Energy production, exploration, distribution, equipment, and related services.
Utilities	Electricity, natural gas, water, and other utility services.
Materials	Chemicals, metals, mining, construction materials, and other raw materials.
Real Estate	Real estate ownership, development, management, and related services, including REITs.
Crypto	Digital assets using blockchain technology rather than representing traditional equity sectors.

9 Formulas & Functions

9.1 Formulas

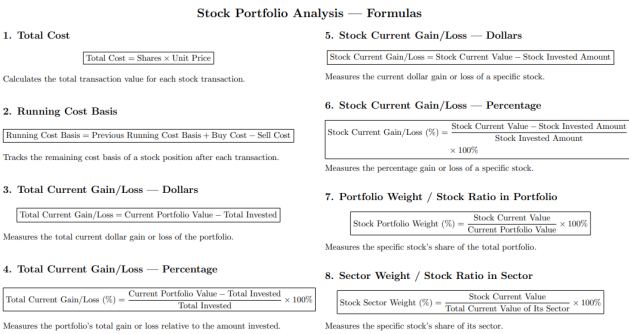


Figure 5: Formulas

9.2 Functions

Function	Purpose
IF	Returns one value when a condition is TRUE and another value when it is FALSE.
IFERROR	Returns a specified value when a formula results in an error.
LET	Assigns names to values or intermediate calculations within a formula.
FILTER	Filters a range or array based on specified criteria.
SCAN	Performs a calculation sequentially across an array and returns each intermediate result.
SEQUENCE	Generates a sequence of consecutive numbers in an array.
LAMBDA	Creates custom reusable functions using defined parameters and calculation logic.
SPLIT	Divides text into separate values based on a specified delimiter.
VALUE	Converts text that represents a number into a numeric value.
INDEX	Returns a value from a specified position within a range or array.
ROWS	Returns the number of rows in a range or array.
MAP	Applies a specified calculation to each value in an array.
UNIQUE	Returns a list of unique values from a range or array.
SUM	Adds numbers together and returns their total.

SUMPRODUCT	Multiplies corresponding values in arrays and returns the sum of the resulting products.
MINIFS	Returns the smallest value that meets one or more specified criteria.
TODAY	Returns the current date.
DATEDIF	Calculates the difference between two dates in specified units.
SORT	Sorts the values in a range or array based on specified criteria.
XLOOKUP	Searches for a value and returns the corresponding value from another range or array.
MAX	Returns the largest value in a range or array.
GOOGLEFINANCE	Retrieves financial market and security information from Google Finance.
SUMIF	Adds values that meet a specified condition.
MAXIFS	Returns the largest value that meets one or more specified criteria.

9.3 Function Logistics

9.3.1 Function 01: Running Cost Basis

```
=IF(B2="", "",
LET(
t, B2,
shares, FILTER($C$2:C2, $B$2:B2=t),
prices, FILTER($D$2:D2, $B$2:B2=t),
states, SCAN("0|0", SEQUENCE(ROWS(shares))),
LAMBDA(acc, i,
LET(
x, SPLIT(acc, "|"),
oldShares, VALUE(INDEX(x, 1, 1)),
oldBasis, VALUE(INDEX(x, 1, 2)),
s, INDEX(shares, i),
p, INDEX(prices, i),
newShares, oldShares+s,
newBasis,
IF(
s>0,
oldBasis+s*p,
IF(
newShares<=0,
0,
oldBasis+s*(oldBasis/oldShares)
)
),
newShares&"|"&newBasis
))),
last, SPLIT(INDEX(states, ROWS(states)), "|"),
VALUE(INDEX(last, 1, 2))
))
```

Purpose

Calculate the running total cost basis for each stock ticker by tracking purchases and sales sequentially.

Logic

1. Identify the current Ticker — LET.

Stores the ticker from B2 as variable *t*.

2. **Retrieve transactions for the Ticker — FILTER.**
Extracts the relevant Shares and Unit Price records for the current ticker.
3. **Create transaction sequence — ROWS + SEQUENCE.**
Counts filtered transactions and generates a sequential index for each one.
4. **Process transactions sequentially — SCAN + LAMBDA.**
Calculates running Shares and Cost Basis while carrying the previous result forward.
5. **Retrieve previous Shares and Cost Basis — SPLIT + INDEX + VALUE.**
Separates previous running values and converts them into numbers.
6. **Retrieve current Shares and Unit Price — INDEX.**
Gets the Shares and Price for the current transaction.
7. **Calculate new Shares — +.**
Adds current transaction Shares to previous Shares.
8. **Handle Buy vs. Sell — IF.**
Determines whether to add purchase cost or reduce Cost Basis.
9. **Calculate Buy Cost Basis — + and *.**
Adds $\text{Shares} \times \text{Unit Price}$ to previous Cost Basis.
10. **Calculate Sell Cost Basis — /, +, *.**
Uses current average cost per share to reduce Cost Basis.
11. **Reset fully sold positions — IF.**
Sets Cost Basis to 0 when remaining Shares are ≤ 0 .
12. **Store the running result — &.**
Combines Shares and Cost Basis into one value for the next SCAN iteration.
13. **Return the latest Cost Basis — INDEX + SPLIT + VALUE.**
Retrieves the final running state and returns the latest Cost Basis.

In Short

- LET $\rightarrow$ identify & organize variables
- FILTER $\rightarrow$ retrieve ticker transactions
- ROWS / SEQUENCE $\rightarrow$ create transaction sequence
- SCAN / LAMBDA $\rightarrow$ process transactions sequentially
- SPLIT / INDEX / VALUE $\rightarrow$ retrieve running values
- IF $\rightarrow$ handle Buy / Sell / fully sold
- + / * / $\div$ $\rightarrow$ calculate Shares & Cost Basis
- & $\rightarrow$ carry the running state
- INDEX / SPLIT / VALUE $\rightarrow$ return final Cost Basis

9.3.2 Function 02: Total Invested

```
=SUM(MAP(UNIQUE(FILTER('Transection History'!B2:B,'Transection History'!B2:B<>"")),LAMBDA(t,
INDEX(FILTER('Transection History'!G2:G,'Transection History'!B2:B=t),ROWS(FILTER('
Transection History'!G2:G,'Transection History'!B2:B=t))))))
```

Purpose

Calculates the total invested amount by summing the latest total cost for each unique stock ticker.

Logic

1. **Identify unique Tickers — FILTER + UNIQUE.**
Retrieves all non-blank tickers and creates a unique list.
2. **Process each Ticker — MAP + LAMBDA.**
Iterates through each unique ticker individually.
3. **Retrieve all Total Cost values — FILTER.**
Filters the Total Cost column for the current ticker.
4. **Identify the latest transaction — ROWS + INDEX.**
Uses the number of filtered transactions to retrieve the last Total Cost value.
5. **Sum the latest value for each Ticker — SUM.**
Adds the latest Total Cost from every unique ticker to calculate Total Invested.

In Short

- FILTER / UNIQUE → identify unique tickers
- MAP / LAMBDA → process each ticker
- FILTER → retrieve transaction costs
- ROWS / INDEX → get the latest Total Cost
- SUM → calculate Total Invested

9.3.3 Function 03: Amount Invested

```
=IFERROR(
LET(
sector,L26,
tickers,UNIQUE(
FILTER(
'Transection History'!B$2:B,
'Transection History'!F$2:F=sector,
'Transection History'!B$2:B<>"
)
),
tickerCosts,MAP(
tickers,
LAMBDA(ticker,
LET(
shares,FILTER('Transection History'!C$2:C998,'Transection History'!B$2:B=ticker),
```

```

costs,FILTER('Transection History'!E$2:E998,'Transection History'!B$2:B=ticker),
n,ROWS(shares),
idx,SEQUENCE(n),
runningShares,SCAN(0,idx,LAMBDA(acc,i,acc+INDEX(shares,i))),
runningCost,SCAN(0,idx,LAMBDA(acc,i,
LET(
prevShares,IF(i=1,0,INDEX(runningShares,i-1)),
q,INDEX(shares,i),
cost,INDEX(costs,i),
IF(
q>0,
acc+cost,
IF(
prevShares=0,
0,
acc+q*(acc/prevShares)
)
)
))),
INDEX(runningCost,n)
)
),
SUM(tickerCosts)
),
0)

```

Purpose

Calculates the total amount invested for a selected sector by determining the remaining cost basis of each stock in that sector and summing them.

Logic

1. **Identify the selected Sector — LET.**
Stores the sector from L26 as sector.
2. **Identify unique Tickers within the Sector — FILTER + UNIQUE.**
Filters Transaction History by sector and removes blank tickers and duplicates.
3. **Process each Ticker separately — MAP + LAMBDA.**
Calculates the invested amount for each stock in the sector.
4. **Retrieve each Ticker's transactions — FILTER.**
Retrieves its Shares and transaction Costs.
5. **Create transaction sequence — ROWS + SEQUENCE.**
Determines the number and order of transactions.
6. **Calculate Running Shares — SCAN + LAMBDA + INDEX.**
Sequentially adds each transaction's Shares to determine remaining Shares Owned.
7. **Calculate Running Cost Basis — SCAN + LAMBDA + IF + INDEX.**
Sequentially updates Cost Basis based on each Buy or Sell transaction.
8. **Handle Buy transactions — IF + +.**
Adds transaction Cost to previous Cost Basis.

9. **Handle Sell transactions — IF + / + *.**
Reduces Cost Basis using current average cost per share.
10. **Handle fully sold positions — IF.**
Sets Cost Basis to 0 when previous Shares Owned are 0.
11. **Return each Ticker's final Cost Basis — INDEX.**
Retrieves the last Running Cost Basis for each ticker.
12. **Calculate Sector Amount Invested — SUM.**
Adds the final Cost Basis of all tickers in the selected sector.
13. **Handle calculation errors — IFERROR.**
Returns 0 if the calculation produces an error.

In Short

- LET → identify Sector
- FILTER / UNIQUE → identify Sector's unique Tickers
- MAP / LAMBDA → calculate each Ticker
- FILTER → retrieve transactions
- ROWS / SEQUENCE → create transaction sequence
- SCAN / LAMBDA → calculate Running Shares & Cost Basis
- IF → handle Buy / Sell / fully sold positions
- INDEX → retrieve transaction & final values
- SUM → calculate total Amount Invested
- IFERROR → return 0 if an error occurs

9.3.4 Function 04: Real Time Value

```
=IFERROR(SUMPRODUCT(('Real Time Market Price'!$G$2:$G998=L26)*('Real Time Market Price'!$C$2:$C998)*('Real Time Market Price'!$B$2:$B)),0)
```

Purpose

Calculates the current market value of all stocks within the selected sector using real-time market prices and current holdings.

Logic

1. **Identify the selected Sector — L26.**
Uses the sector listed in L26 as the filtering criterion.
2. **Filter stocks by Sector — =.**
Checks whether each stock's sector matches the selected sector.
3. **Retrieve Current Price — Column C.**
Uses the current market price for each matching stock.

4. Retrieve Shares Owned — Column B.

Uses the current shares held for each matching stock.

5. Calculate each Stock's Current Value — *.

Multiplies Current Price $\times$ Shares Owned.

6. Calculate Sector Real Time Value — SUMPRODUCT.

Sums the Current Value of all stocks belonging to the selected sector.

7. Handle calculation errors — IFERROR.

Returns 0 if the calculation produces an error.

In Short

- = $\rightarrow$ identify stocks in the selected sector
- Column C $\times$ Column B $\rightarrow$ calculate each stock's current value
- SUMPRODUCT $\rightarrow$ total the sector's real-time value
- IFERROR $\rightarrow$ return 0 if an error occurs

9.3.5 Function 05: Purchase History Tracker

```
=SORT(
  FILTER(
    'Transaction History'!A2:G1000,
    'Transaction History'!B2:B=$B$1
  ),
  1,
  True
)
```

Purpose

Displays the complete purchase and transaction history for the selected stock ticker, sorted chronologically by transaction date.

Logic**1. Identify the selected Ticker — \$B\$1.**

Uses the ticker selected in cell B1 as the filter criteria.

2. Filter transactions by Ticker — FILTER.

Retrieves all transaction records from columns A:G where the Ticker matches \$B\$1.

3. Sort transactions by Date — SORT.

Sorts the filtered transaction records based on the first column, which contains the transaction date.

4. Sort in ascending order — TRUE.

Arranges transactions from the earliest date to the latest date.

In Short

- \$B\$1 $\rightarrow$ identify selected Ticker

- FILTER → retrieve matching transaction history
- SORT → organize transactions chronologically
- TRUE → sort from earliest to latest